

Image Classification via AI

Tech: Python, UV, Streamlit, Tensorflow, Numpy, OpenCV, & PIL.

ML Algorithm: MobileNetV2, a light weight CNN is used to classify the images. From Tensorflow Keras API.

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image.jpg 10.9KB



Uploaded Image



Classify Image

image.jpg 10.9KB



Uploaded Image

 Classify Image

Predictions

sports_car: 44.880766%

beach_wagon: 3.759195%

convertible: 3.519082%



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image 2.jpg 0.6MB



Uploaded Image



Classify Image



Uploaded Image

 Classify Image

Predictions

lakeside: 19.123589%

promontory: 15.956214%

fountain: 15.630576%

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image 3.jpg 10.7KB



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Classify Image



Uploaded Image

 Classify Image

Predictions

drilling_platform: 12.062261%

mosque: 4.599381%

dome: 4.537615%

File Edit Selection View Go Run Terminal Help

image_classifier

EXPLORER

image_classifier.py U main.py 2, U X

main.py > ...
1 # Author: Kirujan Je / December 2025
2 # Tech: Python, UV, Streamlit, Tensorflow, Numpy, OpenCV, & PIL.
3 # ML Algorithm: MobileNetV2, a light weight CNN is used to classify the images. From Tensorflow Keras API.
4
5 import cv2 #opencv
6 import numpy as np #installed within tensorflow as default
7 import streamlit as st
8 from tensorflow.keras.applications.mobilenet_v2 import (
9 MobileNetV2,
10 preprocess_input,
11 decode_predictions
12)
13 from PIL import Image #image library in python, in cv2 as default
14
15 # mobilenet_v2 is a popular pretrained model in the tensorflow library,
16 # It is a lightweight model, suitable to use on a laptop
17 # this avoids having to train our own model, and a significant amount of time

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:From C:\Users\kiruj\OneDrive\Desktop\image_classifier\.venv\Lib\site-packages\keras\src\backend\common\global_state.py:82: The name tf.reset_d
efault_graph is deprecated. Please use tf.compat.v1.reset_default_graph instead.

1/1 2s 2s/step
1/1 1s 1s/step
1/1 1s 842ms/step
1/1 1s 834ms/step
[]

uv
uv
uv

Ln 12, Col 2 Spaces: 4 UTF-8 LF Python 3.12.10 (Microsoft Store)